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PERSPECTIVE

Neuroscientific insights about computer vision models: a concise review

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Abstract
The development of biologically-inspired computational models has been the focus of study ever since the artificial neuron was introduced by McCulloch and Pitts in 1943. However, a scrutiny of literature reveals that most attempts to replicate the highly efficient and complex biological visual system have been futile or have met with limited success. The recent state-of the-art computer vision models, such as pre-trained deep neural networks and vision transformers, may not be biologically inspired per se. Nevertheless, certain aspects of biological vision are still found embedded, knowingly or unknowingly, in the architecture and functioning of these models. This paper explores several principles related to visual neuroscience and the biological visual pathway that resonate, in some manner, in the architectural design and functioning of contemporary computer vision models. The findings of this survey can provide useful insights for building futuristic bio-inspired computer vision models. The survey is conducted from a historical perspective, tracing the biological connections of computer vision models starting with the basic artificial neuron to modern technologies such as deep convolutional neural network (CNN) and spiking neural networks (SNN). One spotlight of the survey is a discussion on biologically plausible neural networks and bio-inspired unsupervised learning mechanisms adapted for computer vision tasks in recent times.

Keywords Biological Vision · Computer vision · Deep learning. Spiking neural networks · Biologically plausible neural networks

1 Introduction

Brain-inspired computational models were introduced way back in the 1940s (McCulloch and Pitts 1943; Hebb 1949) in the form of feedforward Artificial Neural Network (ANN). The actual human brain comprises of billions of neurons interconnected by trillions of synapses (Herculano-Houzel 2012). It is the connectivity and parallelism of the neurons that makes up for the sluggish performance of a single neuron (Cox and Dean 2014), by propagating information between distant regions of the brain in a few synaptic steps. The exact simulation of the complicated network and interconnections in the brain is still under investigation, though attempts have been made to simulate parts of the computational process.

The first artificial neuron was introduced in 1943 by McCulloch and Pitts (1943), based on Alan Turing's idea of an atomic computational unit (Turing 1937). Rosenblatt (1958) introduced the perceptron which is the basic decision-making unit of the current version of ANN. The perceptron can be stated to be the first simulation of a biological neuron that is triggered on receiving a significant-enough stimulus from a sensory input or other neurons, causing a spike in the membrane potential from the resting state (OFF state) to the action potential (ON state) as illustrated in Fig. 1a. Likewise, the artificial neuron or perceptron is engineered to switch between two states using a non-linear function called the activation function. The artificial neuron implements a linear summation of the inputs followed by a non-linear activation function that translates the summation of inputs into a binary output as shown in Fig. 1b.

Minsky and Papert (1969) introduced the concept of parallel computations within a network of perceptrons; they inserted a hidden layer in between the input and output layers that facilitated non-linear mapping. The connection weights

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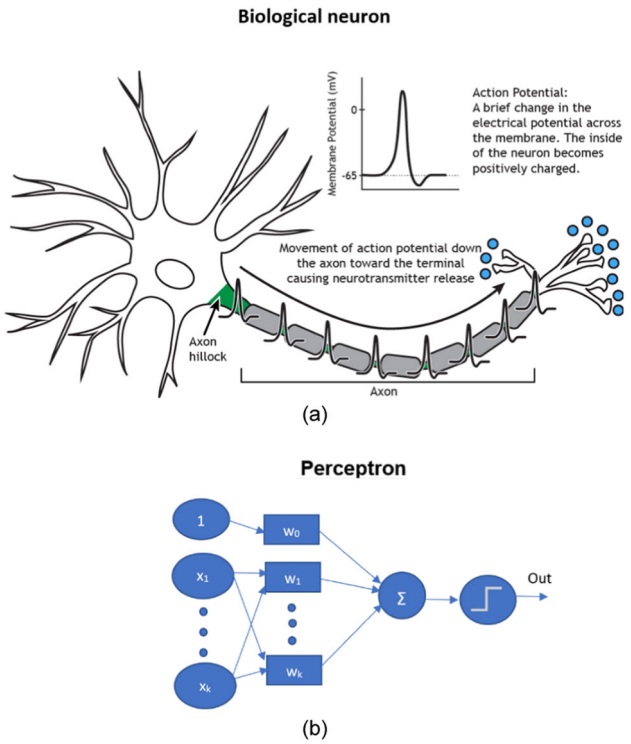

Fig. 1 **a** Biological neuron and its excitation shown as a change from the resting membrane potential to the action potential (Adapted from Henley (2021)) **b** Perceptron or artificial neuron: simulation of the biological neuron

became, subsequently, a matter of discussion. Several optimization algorithms were proposed to determine the network parameters, such as the backpropagation algorithm (Rumelhart et al. 1985) based on the stochastic gradient descent, that is used successfully to train the state-of-the-art deep neural networks today. Its inspiration from biology is though not very clear, since neurons in the brain are not known to propagate errors backward through multiple synapses (Lillicrap et al. 2020). The brain does, however, update the synaptic connections between neurons on a regular basis. Pyramidal neurons integrate information from the feedforward and feedback pathways (Johnson and Burkhalter 1996; Vezoli et al. 2021). While the forward pathway provides "ascending" information about the objects or the other details in a scene, the feedback pathway provides "descending" information about the context in which a stimulus appears.

Research in neural networks that process visual information got a boost in the late 1990s with the introduction of LeCun et al.'s Convolutional Neural Network (CNN) (LeCun et al. 1995, 1998) that processes an image using smaller-sized convolutional filters, and then pools the filtered outputs across multiple regions. Biologically-inspired modifications of the CNN architecture have been introduced. Some of these, such as LGN-CNN, will be discussed in this survey. Over the years the CNN architecture has evolved into more sophisticated structures with several hidden layers in the order of 100 to 200, an example being the DenseNet-201 architecture introduced in 2017 (Huang et al. 2017), with the singular goal to achieve human-like performance in recognition and prediction. Researchers, in today's times, develop deep learning architectures and learning mechanisms without gaining inspiration from biology, by specifying the architecture, hyperparameters, learning rate, and the objective function that is used to ascertain how far the neural network is from achieving its goal.

Though the biological connect with evolving neural network architectures was broken, there have always been groups of researchers who have worked continually towards understanding the biological visual systems found in different species (Hubel and Wiesel 1959; Eckhorn 1999; Wurtz 1969; Nakada et al. 2021). The efforts of these researchers have, over the years, paved the way for the development of biologically plausible neural networks and learning mechanisms for the development of brain-inspired intelligence and visual perception (BIVP) (Wang et al. 2020a, b; Yang et al. 2022). A few notable studies are mentioned below.

A survey published in 2019 by Li et al. (2019) promotes the integration of neuroscience and robotics to create robots with human-like brains. Extending this idea, Chen et al. (2023) recently proposed the use of biological neural networks (BNN) as robot brains. The idea was to use biological intelligence to control the movement of robots. Several BRAIN projects were initiated that seek to unravel the mystery of the working of the living brain (Amunts et al. 2019). The BLUE BRAIN project is notable for attempting to simulate the brains of living organisms on IBM's Blue Gene supercomputer (Markram 2006). Neuromorphic chips are recent advancements in emulating brain functionalities that have brain-inspired hardware and can used in Brain-Machine Interfaces (BMI) for the treatment of neurological disorders such as epilepsy (Yoo et al. 2021). These chips are fast and energy-efficient since they do not rely on the von Neumann architecture.

A notable survey related to the current study was conducted a decade ago by Cox and Dean (2014), that investigates the interplay of visual neuroscience, computer vision and machine learning for understanding scene content and recognizing objects. They defined brains as computing machines that collate multimodal and ambiguous information and transform these into coherent thought and action. Medathati et al. (2016) conducted a review of literature and suggested a synergy between computer vision and biological vision models for the specific tasks of image sensing, scene segmentation and optical flow. Most of the models reviewed advocated some form of integration between the low-level (sensory) and high-level (semantic) visual information. Kreiman (2021) adopted a neuro-scientific approach to

study the computations in the visual cortex. Several modern computer vision tools such as Deep CNNs were reviewed. The author concluded that the simulation of vision is an unsolved problem. Cohen et al. (2022) review cognitive neuroscience principles that may help to refine ANN architectures. The authors suggest modification of the recurrent neural network (RNN) architecture for dynamic and flexible application of decision rules in order to model complex cortical responses. Recent developments in computer vision include biologically plausible neural networks that operate on self-supervised learning (Tang et al. 2022). The aim is to use unlabeled samples for training the neural network instead of using class labels. Unsupervised models that operate with self-supervision in the absence of labelled data are now regarded as the new face of future computer vision models.

These topics and many more will be covered in this survey with the aim of understanding the progress achieved and identifying potential research directions for the development of bio-inspired computer vision model architectures and learning mechanisms. The contributions of this work are as follows: (i) explores the biological connection of the state-of-the-art computer vision models (ii) surveys recent advancements in biologically-plausible neural networks and learning mechanisms that have been adapted for computer vision (iii) emphasizes on the role of biological vision in influencing future research in computer vision (iv) outlines several open research problems, and challenges involved in embedding neuroscientific principles into computer vision models. The organization of the rest of this paper is as follows. Section 2 contains a detailed discussion on the state-of-the-art computer vision models and their biological connection. Section 3 presents the recent developments on biologically plausible neural networks which includes spiking neural networks, bio-inspired CNNs, and bio-inspired unsupervised learning mechanisms. The research problems, challenges and future directions are discussed in Sect. 4, and the final conclusions are drawn in Sect. 5.

2 Biological insight into traditional state-of-the-art computer vision models

This section reviews several traditional state-of-the-art computer vision models that have been in popular use over the years and analyze their biological connection or inspiration, either in terms of the architecture or the learning mechanism.

2.1 Multi-layer perceptron (MLP)—the primitive ANN model

The Multi-Layer Perceptron (MLP) artificial neural network is the most primitive computer vision model that accepts at its input the linearized or flattened input image. Predicting the category of the flattened image via ANNs is well-explored research (Agrawal and Agrawal 2015; Marinai et al. 2005). In the deep learning jargon, the fully connected layer at the output of all deep CNNs is a MLP that learns linearized feature maps derived from the last convolutional layer. A neuron in a fully connected layer is connected to all other neurons in the next layer. The biological neuron, however, has more functionalities (Trobe 2001). This was one of the drawbacks listed by computational neuroscientists in their objection to consider the ANN as a computational unit of the biological brain (Shao and Shen 2023). The artificial neurons are connected to the next layer by a feedforward connection, and the artificial neural network is trained using the backpropagation algorithm. There is no evidence of backpropagation in the biological neural structure (Lillicrap et al. 2020; Solomon and Rosa 2014). Biologically-plausible backpropagation is ongoing research (Liu et al. 2022); more details will be covered in Sect. 3. The visual processing in the cortex is much more intricate due to the many recurrent and feedback connections between sub-cortical areas (Cox and Dean 2014). There is, however, an important similarity between ANN and the brain neural network. The set of connection weights in both cases is the outcome of learned associations (Kreiman 2021). The strength of the connection implies more weight to a particular input. When the number of hidden layers in an ANN is increased to three or more, the network becomes "deep" giving rise to the term deep neural networks (DNN) or deep nets (Kriegeskorte 2015).

2.2 Neocognitron—the precursor to CNNs

ANNs that dominated the field till 1980s required a linearized input, and did not accept a two-dimensional image as the input. The scenario shifted significantly when Fukushima (1980) introduced the neocognitron for visual pattern recognition, which is a neural network facilitating hierarchical signal processing comprising of alternating layers of S-cells and C-cells. While S-cells have smaller receptive fields, the C-cells have larger receptive fields. The neocognitron architecture is inspired from the hierarchical arrangement of the simple cells or lower-order hyper-complex cells in the primary visual cortex and the higher-order complex cells in the V2, V4 cortical regions. The neocognitron accepts a two-dimensional image as the input and hierarchically processes it via alternate layers of S-cells and C-cells, extracting local features in the initial stages and global features in the higher layers; the last C-cell layer being the recognition layer. While the S-cells, having smaller receptive fields, are responsible for feature extraction, the C-cells, having larger receptive fields, pool the outputs of S-cells connected at their input. The learning algorithm of the neocognitron is based on self-organization wherein the S-cells have their input connections

strengthened in response to a specific input feature. Each C-cell receive its input from a fixed set of S-Cells and records the strongest response at its input.

2.3 Convolutional neural network (CNN)

The neocognitron is regarded as the precursor to LeCun's CNN, also known as ConvNet, which forms the backbone of the current state-of-the-art computer vision models. As shown in Fig. 2, CNN comprises of cascaded stages of convolutional and pooling layers that compute a hierarchical feature representation from an input image (LeCun et al. 1995, 1998; Krizhevskyet al. 2012). Each convolutional layer is followed by a sub-sampling procedure such as max pooling or average pooling, and a non-linear activation function such as the Rectified Linear Unit (ReLU) activation function. At each intermediate layer, new features are computed that are known as activation maps. These features range from the simple (initial layers) to the complex (higher or deeper layers). The last few layers comprise of fully connected or dense layers followed by a probabilistic softmax classification layer. Figure 2 shows a generic CNN architecture with three cascaded convolutional (and pooling) layers, two fully-connected layers and a softmax layer, that resembles the cascaded stages of S-Cell and C-cell layers of a neocognitron.

The CNN proved to be a precursor to different kinds of deep neural networks where the researchers focused on increasing the depth of the network to attain human-like accuracy (Kruthiventi al. 2017). Deep CNNs have proved to be powerful high-performance tools in the field of computer vision, achieving human-like performance in visual recognition tasks (Huang et al. 2017; Cadena et al. 2019; Yuan and Lu 2015; Sun et al. 2018). The results reportedly, are more accurate than most of the handcrafted features which may not cover all levels of visual complexities such as the simultaneous effects of multiple-views, time, shadows, sunlight and occlusions (Wirges et al. 2018; Zhi et al. 2018; Susan and Tuteja 2024). The depth of the model indicates the number of layers in the deep framework. Deep CNNs trained on very large-scale datasets constitute the state of the art nowadays. Several comprehensive, large-scale labeled datasets such as ImageNet have been introduced for scene and object recognition, which when trained on high performance GPUs and parallel computing clusters give high recognition performance. Deep CNNs are custom-trained on a specific recognition task unlike the biological brain that can handle multiple categorization tasks simultaneously. Biological vision is known to adapt to unseen objects quickly by drawing inference from the characteristics of known objects (Goodale and Milner 2013). Deep CNN classifiers that learn from few training instances (few-shot/one-shot learning) (Wang et al. 2020a, b; Vinyals et al. 2016) or learn to classify never-seen-before instances (zero-shot learning) (Zhang et al. 2017) are recent developments that strive to emulate the human ability to recognize objects whose samples were never seen before.

The features extracted from intermediate layers of a Deep CNN may be fused with deep features extracted from other networks or modalities; this type of research is popular in deep learning (Song et al. 2018). Sun et al. (2018) appreciated the fact that a single representation is most often not enough to describe an image. They fused three types of deep features:—object semantics derived from deep networks, contextual information derived from long short-term memory model, and global features derived from the activation layer of a CNN. This implies that effective understanding of an image involves the fusion of both global and local information. Early-cum-late fusion of local and global features have been proposed by Susan and Malhotra (2021) as a crude representation of the integration of signals in the biological visual system. A multi-scale ConvNet was proposed in (Farabet et al. 2013) for scene labeling, that was used to derive a multi-scale deep feature representation encompassing texture, shape, and context of a scene. Meaningful components were then selected from a segmentation pool that best explains the contents of the scene. This work propounded the theory that multi-scale deep learning with sufficient postprocessing aids in understanding the scene best. Latest research attempts to incorporate the multi-tasking functionality of the brain into the deep learning environment. There are several examples in literature of parallel deep CNNs frameworks that have been used to combine multi-source inputs for various classification tasks (Palvanov and Cho 2019).

2.4 Biological connect and disconnect of CNNs

2.4.1 Biological connect of CNNs

The biological origins of CNN were linked to the simple and complex cell theory of Hubel and Wiesel by its creators (LeCun et al. 1998). In subsequent years, researchers (Cox and Dean 2014; Shafi 2018; Lindsay 2021) conducted further studies to draw parallels between CNN and the biological visual system. This sub-section examines different architectural components and functional behavior of CNNs to highlight the biological comparisons. Lindsay (2021) equated the filtering, pooling and rectifying operations in CNNs to individual computations in each of the cortical regions V1, V2, V4 and ITC that produce their own individual feature maps. There are two types of pooling operations employed in CNNs – average which is a weighted linear summation of inputs with all inputs having equal weights, and max-pooling. Physiological experiments in (Riesenhuber and Poggio 1999) established that a non-linear maximum operation was a better approximation to the position-invariant pooling performed by a complex cell in the primary visual cortex than

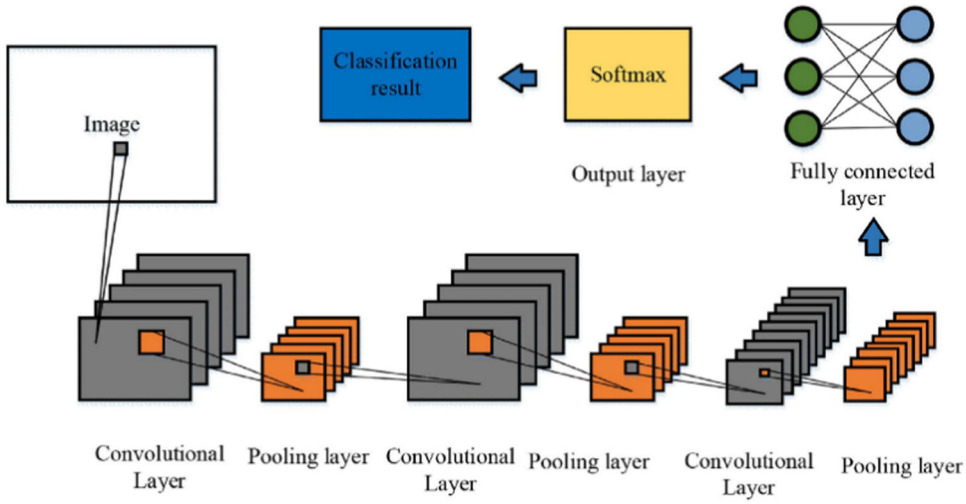

Fig. 2 A general CNN architecture comprising of a series of convolutional layers and pooling layers followed by one or more fully connected layers and a softmax layer (Adapted from Liu (2021))

the linear summation of inputs. The max-pooling operation records the strongest response among the input stimuli of the complex cell comprising an array of simple cells. However, max-pooling does not generalize well to unseen data and may fail for noisy images (Akhtar and Ragavendran 2020). Biologically-inspired pooling operations have been proposed as alternatives that suggest better regularization than max-pooling (Hyvärinen and Köster 2007). The rectifying operation performed using the *tanh*, sigmoid, ReLU or leaky ReLU function in CNN layers mimics the non-linear neuronal activations in biological neurons.

The most significant biological resemblance of CNNs lies in the hierarchical visual information processing that occurs across the successive layers of the CNN, akin to the hierarchy existing among the visual cortex regions along the biological visual pathway (Cox and Dean 2014; DiCarlo et al. 2012; Huang et al. 2012; Shafi 2018; Cichy et al. 2016). Kriegeskorte (2015) discovered an anatomical analogy between the response of a trained CNN to an unknown image and the brain of the Rhesus monkey that was shown an unknown image. The signals at several places in the ventral visual stream were found to be matching with the activations of different CNN layers. Another notable research is that of Cichy et al. (2016) who drew attention to the similar spatio-temporal hierarchical relationship between CNN and the early areas of the ventral and dorsal visual streams. A correspondence between the hierarchical signal processing in the visual pathway and CNNs is established by Cox and Dean (2014) and Xu and Vaziri-Pashkam (2021), highlighting the aggregation of visual cues as the signals traverse from the lower end of the hierarchy to the higher end. Figure 3a illustrates the ventral and dorsal visual pathways found in the brain. While the ventral visual pathway is responsible for recognizing faces, objects and scenes, the dorsal visual stream is responsible for the spatial arrangement of objects in the scene and object-guided motion. The visual cortex is divided into five cortical areas V1-V5 based on the function and structure (Huff et al. 2018). Figure 3b reveals a hierarchical flow of information along the LGN-V1-V2-V4-ITC stages in the ventral visual pathway, as found in CNNs, however, with additional direct or lateral recurrent connections between different areas. In the ventral visual pathway, the visual signal from the retina reaches the V1 area or the primary visual cortex via the thalamus, and the V2 and V4 cortical areas before reaching the Inferior Temporal (IT) Cortex. The initial stages of the ventral stream detect edges, orientations, lines and colors. The latter stages recognize places, objects, and faces. While V1 performs spatial frequency filtering of the visual information received from the LGN (De Valois et al. 1982), V2 detects contours, colors and object orientation (Peterhans and Heydt 1989), V4 is sensitive to shape orientation and curvature and depth, and the ITC detects objects, faces and scenes. (Desimone et al. 1984; Trobe 2001; Xu and Vaziri-Pashkam 2021). Cadena et al. (2019) tested two learning paradigms i.e. transfer learning vs. data-driven end-to-end training of deep CNNs for predicting the spiking activity in V1 in response to natural images. The authors found that both approaches performed reasonably well though the transfer learning using pre-trained models involved less experimental time.

2.4.2 Biological disconnect of CNNs

As observed from Fig. 3b, one primary difference between CNNs and biological vision lies in the complex feedback, recurrent and parallel connections existing between the cortical areas in the biological visual system that facilitates to and fro flow of information in the higher visual areas. CNN exhibits, on the other hand, a feedforward flow of information. The presence of reciprocal and parallel connections in between the cortical regions was reported by several researchers (Trobe 2001; Rockland and Pandya 1979; Shapley 1990; Felleman and Van Essen 1991. A notable

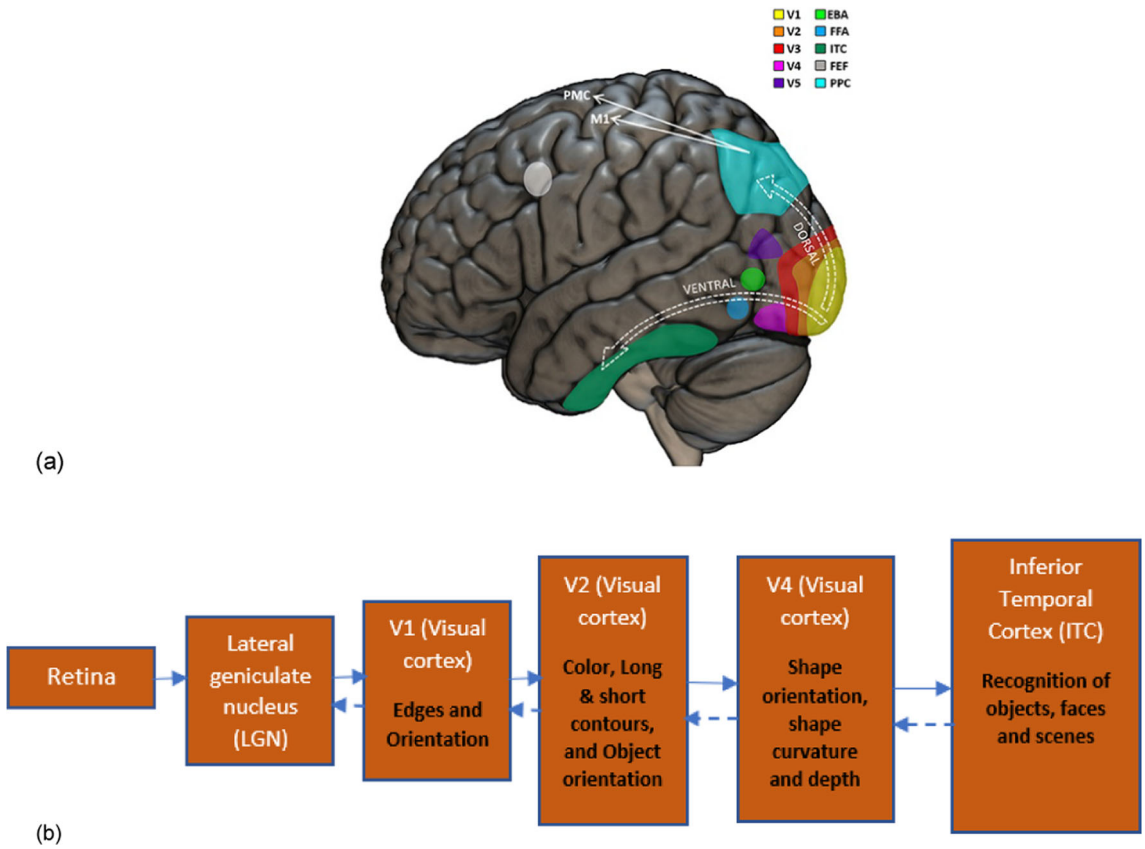

Fig. 3 **a** The dorsal and ventral visual pathways through the visual cortex areas—V1, V2, V3, V4 and V5, extrastriate body area (EBA), fusiform face area (FFA), inferior temporal cortex (ITC), frontal eye field (FEF), posterior parietal cortex (PPC), primary motor cortex (M1) and premotor cortex (PMC) (Adapted from Ionta (2021)), **b** The hierarchical stages along the ventral visual pathway with parallels drawn to popular computer vision tasks

study is that of Felleman and Van Essen (1991) who identified interwined reciprocal connections within a hierarchical organization of the brain cortical regions, ranging from the low-level V1, V2 cortical regions to the high-level V3, V4 and ITC cortical regions, and somatosensory/motor areas lying outside the visual cortex. As observed from the dotted reverse arrows in between the blocks in Fig. 3b, there are local feedback connections between the LGN, V1, V2, V4 and IT cortical regions in the biological visual system (Cox and Dean 2014), unlike the feedforward flow of information in CNNs. It is a well-known fact that CNNs learn their weights through backpropagation, like ANNs (Goodfellow et al. 2016), hence the learning algorithm is certainly not biologically motivated (Lillicrap et al. 2020; Solomon and Rosa 2014). Biologically plausible alternatives to the backpropagation algorithm will be discussed in Sect. 3.

2.5 Deep neural networks incorporating attention mechanism

Attention mechanisms optimize the performance of visual systems and guides the observer's eyes towards salient features or locations in the image. There are two broad classifications of visual attention (Lindsay 2020; Carrasco 2011). The first classification based on the approach followed is (i) top-down attention driven by prior information which shifts an observer's attention to where the target is likely to be (ii) bottom-up attention driven by stimulus which shifts an observer's attention to where a salient visual feature is located. The second classification based on the guidance factor is (i) spatial attention which is guided by a particular location (ii) feature-based attention that is guided by salient features such as color, orientation and motion direction (iii) object-based attention which is guided by object structure and shape. While spatial and object-based attention mechanisms follow a top-down approach, feature-based attention mechanisms follow a bottom-up approach. Pan et al. (2023) combine bottom-up and top-down attention mechanisms for

image captioning. While bottom-up attention mechanism detects all the objects in the image, the top-down attention mechanism creates a sequence of selected objects that is used for caption generation. A bottom-up attention mechanism was employed in (Adámek et al. 2024) to detect abnormalities in the gaze of schizophrenia patients for highly salient images, the saliency being measured by physical attributes such as color, brightness, orientation and motion. Novel deep learning architectures inspired by the human visual attention mechanism have been recently introduced, such as DeepFix that predicts eye fixations by learning a saliency map of multiple scales in a hierarchical manner (Kruthiventi et al. 2017). This agrees with previous anatomical and physiological findings that sensory information is present at multiple levels in the visual cortex which is then integrated in a hierarchical manner through reciprocal connections between cortical regions (Johnson and Burkhalter 1996; Vezoli et al. 2021). Two pre-trained networks, based on AlexNet, were used; one network was trained on ImageNet[1] (objects) and the other on Places205 (scenes). In order to simulate human-like judgement of images, Yuan and Lu (2015) implemented a manifold regularized deep architecture that exploits the structural information by mapping between the visible layer and hidden layer, and learns high-level features in an unsupervised manner. This type of research addresses the problem of bridging the gap between low-level features and high-level semantics.

The latest advancement in attention-based computer vision is the vision transformer (ViT) that was proposed for large scale image recognition (Dosovitskiy et al. 2021). ViT learns from sequential image patches by aggregating information from local and global heads through the self-attention mechanism, thereby performing well for image classification tasks provided that adequate training data is available (Maurício et al. 2023). ViTs are not biologically inspired per se. They were simply adopted from natural language processing to the image domain. Comparison can be drawn between the multi-head self-attention mechanism of ViT and human visual attention that focuses attention on regions of interest in an image, though it is noted that human visual attention is additionally cognizant of the inter-area recurrent processes as well (Freeman et al. 2011). Most recently, researchers have tried to infuse biologically-inspired modifications into the transformer architecture and functioning. More specifically, Tang et al. (2023) introduced a positional embedding mechanism that is reflective of the manner in which the dorsal visual pathway represents the spatial relationships between the objects in a scene. The authors claim that a combination of convolutional layers to capture local features and transformers to capture global features will capture the visual information transfer between the dorsal and ventral visual streams. It is expected that many such bio-inspired variants of transformer-based architectures will emerge in the near future, that is expected to boost the accuracies of computer vision to even higher levels.

[1] https://www.image-net.org/

3 Development of biologically plausible neural networks

Neural networks, like the human brain, learn from experience. This is the basis for most of the analogy studies conducted by enthusiasts seeking to link neural networks with biology. Saxe et al. (2021) attempt to provide evidence of equivalence between multi-task learning procedures in artificial neural networks and biological neural networks. The authors suggest a context-dependent learning mechanism by which there is minimal representational overlap between consecutive learning tasks. Cortical areas comprise of several underlying layers (Vanni et al. 2020) that justify increasing the depth of artificial neural networks. Deep learning by supervision is, however, regarded as biologically implausible by the majority of researchers since the error-driven adjustment of network weights by backpropagation of errors is biologically unfounded (Rolls 2021; Ororbia et al. 2024). Another objection is the existence of complex recurrent or feedback connections between cortical regions in the brain that does not justify the feedback connections symmetric with feedforward connections used in backpropagation (Shervani-Tabar and Rosenbaum 2023). Furthermore, learning under supervision from thousands of examples, as is the requirement for present-day deep neural networks, is not biologically plausible since it does not explain how infants learn to detect and recognize objects and faces without supervision. In this context, unsupervised models of computer vision that learn to classify unseen objects by self-supervised learning are regarded as the emerging face of biologically plausible neural networks. Additionally, spiking neural networks that seek to emulate the firing pattern of neurons in the brain are also considered biologically plausible. This section contains a detailed discussion on biologically plausible neural network models developed for computer vision from the perspective of their bio-inspired architectures and learning mechanisms, and highlight their potential in influencing the design of future computer vision models.

3.1 Biologically-inspired CNNs

The design of a CNN architecture involves several factors of concern, such as the depth of the network, the number and size of filters in the convolutional layers, neuron activation function (eg. ReLU and Leaky ReLU), number and placement of pooling layers and batch normalization layers. Zhang et al. (2016) utilized biological guidance from the human

visual cortex for optimizing the hyperparameters of a deep CNN model. The resulting network achieved high accuracies in the image recognition task. Tripp (2019) further extended the idea and designed a cortex-like CNN architecture that had longer skip connections, larger kernel lengths and strides, and differences in sparsity as compared to conventional CNNs. The CNN hyperparameters were optimized to fit neurophysiological data acquired by tract tracing and electrophysiology studies. An ON/OFF ReLu function was proposed in (Kim et al. 2016) as a bio-inspired neuron activation function. This function passes on both positive and negative information onto the next layer, unlike a conventional ReLu function that passes only positive information onto the higher layers of the network. The authors claim that the ON/OFF ReLU is inspired from the ON/OFF visual pathway in the retina (Hartline 1938). There are two ReLU functions (positive, negative) instead of one, and the weighted linear sum is given as input to both the ReLU functions.

The cross-entropy loss objective function used for training deep neural networks is not used in the brain, as per studies conducted by Richards et al. (2019), since it requires the class label for each sensory input. The authors in (Richards et al. 2019) approve a set of objective functions as being ecologically possible. This includes description length objective function used in predictive coding which is the theory of how the brain performs probabilistic inference (Huang and Rao 2011; Spratling 2017), objective functions used in reinforcement learning to maximize rewards (Williams 1992), and increase of mutual information due to reduction of uncertainty during interaction with the environment. The integration of backpropagation with Hebbian-like learning mechanisms such as burst-dependent synaptic plasticity was propounded in Payeur et al. (2021) to coordinate learning in hierarchical circuits. Unsupervised Hebbian rules were put into use for biologically-inspired unsupervised learning way back in 1991 by Földiak and further explored in 1998 (Földiak 1991; Földiak 1998) Transformation-invariant representations of the object were derived by this form of unsupervised learning. The authors demonstrated through simulations and mathematical analysis that a biological neural network with learning rules that are dependent on high frequency bursts combined with synaptic plasticity of feedback pathways, and properties of dendrites and synapses, can perform the same tasks as deep neural networks.

This section also reviews some important CNN architectures that claim to be biologically inspired, primarily due to inclusion of recurrent loops and reciprocal and parallel feedback connections as found in the brain cortical regions. Tang et al. (2022) recently put forth the assertion that enhancing the structural similarities of deep neural networks with the visual cortex in terms of hierarchy of layers and feedback or recurrent connections would help greatly in modeling neural computations. Spoerer et al. (2017) investigated bottom-up, lateral, and top-down connections to emulate the lateral and feedback connections in the biological visual pathway. The resultant architecture was able to recognize objects even in the presence of occlusion. Kubilius et al. (2019) developed CORnet-S which is a shallow artificial neural network with recurrent connections, which the authors claimed to possess brain-like object recognition capability. CORnet-S has four computational areas that approximate the functionalities of the V1, V2, V4 and ITC areas of the ventral stream, followed by a linear category decoder. This work is particularly significant since it propounds the concept of compact recurrent networks instead of the deep feedforward architectures which is the state of the art in computer vision today. Works that integrate nanoscale sensors within the CNN architecture (Porod et al. 2004) to replicate the feedback connections found in the brain are suggestive of interdisciplinary approaches to design of bio-inspired CNNs.

One noteworthy bio-inspired CNN architecture is LGN-CNN (Bertoni et al. 2022) whose first convolutional layer comprises a single filter that mimics the behavior of the Lateral Geniculate Nucleus (LGN) which is the first stage in the visual pathway, followed by a ReLU and a fully connected layer. This layer implements the Laplacian of Gaussian (LoG) which the authors claim to be a good approximation of the behavior of receptive field profiles of the cells of the LGN that exhibits a rotationally symmetric pattern. The second convolutional layer comprising Gabor-like filters mimics the functionalities of the receptive field profiles in the primary visual cortex. Another bio-inspired improvisation in the LGN-CNN architecture is the batch normalization layer that subtracts the mean value from each element to be normalized. The operation is similar to the normalization in the retina in which the mean light intensity is subtracted from photoreceptor responses so that the response of the retina to an image under different lighting conditions would be the same (Carandini and Heeger 2012). The remaining layers of LGN-CNN comprising convolutional, pooling and fully connected layers are, however, not biologically inspired, and function as in a normal CNN. The authors recognize the need for a further investigation in the modifications to be applied to the bio-inspired first two layers while the remaining layers increase in number as happens in deep architectures. The classification performance of CNNs is affected when the input images are corrupted by noise (Dodge and Karam 2017; Kaplan et al. 2021). The authors in (Dapello et al. 2020) provide a solution to this problem by simulating a V1 layer to be used in front of CNNs (network is named as VOneNet) for improved image classification performance that is not affected by image perturbations or noise and is robust to adversarial attacks. Evans et al. (2022) further explored the idea by proposing the use of a biological filter bank approximated by a Gabor filter bank to precede standard deep CNN architectures in order to

reduce the effect of image perturbations. These works suggest that the incorporation of biological constraints improve the performance of deep models and make them resilient to noise, mimicking the human ability to extract information from corrupted images.

3.2 Unsupervised learning mechanisms inspired by the brain

Computer vision models based on bio-inspired unsupervised learning mechanisms have recently emerged as the new face of biologically plausible neural networks that are the closest to the ongoing learning process in the human brain. A Biological Neural Network (BNN) that learns by self-supervision from unlabeled samples is a typical example (Tang et al. 2022). Unsupervised Hebbian rules were used to learn the parameters of a neuro-computational model in (Teichmann et al. 2021) for visual feature codebook generation that can be used for a variety of applications such as object recognition. The development of VisNet, a biologically-plausible four-layered neural network with a hierarchical arrangement of competitive networks is another example (Wallis and Rolls 1997, Rolls et al. 2018). VisNet learns embeddings of invariant views that occur close together in time using a temporal trace rule for view-invariant object recognition. Table 1 showcases some of the latest developments in biologically plausible neural networks that are based on unsupervised learning. The year 2021 saw a boom in the research in this direction, as observed from Table 1. Zhuang et al. (2021) demonstrated that deep unsupervised contrast embedding can be used to train artificial neural networks, and such self-supervised learning mechanisms can achieve the neural prediction accuracy in multiple cortical areas of the ventral visual stream. The authors proved that contrastive learning procedures such as SimCLR with local aggregation outperformed supervised models trained on many examples. More advanced deep neural networks employing self-supervised learning such as CoCoNet (Li et al. 2022) are now introduced for multi-view representations of the object in the ventral and dorsal visual streams. The ventral visual stream performs object recognition while the dorsal visual stream handles motion attributes through optical flow. These are considered as two separate views while the target frame in the video constitutes the third view. The feature representations from multiple views are then shared and mapped to a latent space. A complementarity-aware loss function is then used to train the network. One significant study is that of Geirhos et al. (2021) who proved after an extensive comparative evaluation that self-supervised multimodal models like CLIP that perform language-image training (Radford et al. 2021) perform exceptionally well as compared to supervised CNN models.

3.3 Capsule networks and slot attention modules

Capsule networks or CapsNets were introduced by Sabour et al. (2017) as an alternative to CNNs by replacing the scalar feature detection in CNN layers by the vector output of capsules, and replacing the max-pooling by routing-by-agreement. The dynamic routing prevalent in capsule networks has a biological connect. Hinton, who is one of the authors of capsule networks, had stated way back in 1981 that dynamic connections and frames of references contribute to view-independent shape description as in the biological system (Hinton 1981). The original CapsNet comprises of one convolutional layer and one fully connected capsule layer. It was found to classify MNIST handwritten digits quite well, but was not as efficient in classifying objects from the CIFAR dataset. A deeper version of the of the capsule network was introduced by Rajasegaran et al. (2019) that seemed to resolve the issue. The DeepCaps network, as it was called, incorporated 3D convolution for local dynamic routing and parameter reduction. Further research on going deeper with capsule networks is required to explore more possibilities for improving its object recognition performance. Doerig et al. (2020) explain the significance of recurrent loops in capsule networks for implementing visual grouping and segmentation for shape recognition in a visually crowded scene. This paper, notably, presents psychophysical evidence of recurrent processing in the human visual system, and claims that CapsNets reproduce this behavior well. Capsules are trained to identify specific objects or object parts, which is harmful for generalization, especially when the task involves unseen object discovery. Slot attention modules introduced by Locatello et al. (2020) are even more reliable alternatives to capsule networks, in this regard, since they act as an interface between the perceptual input from CNN and a set of abstract representations of objects called slots whose outputs are integrated for scene cognition. This allows for unsupervised discovery of unseen objects as proved empirically by the authors.

3.4 Spiking neural networks

Only a few techniques like electrophysiology were available in the past to record nerve cell responses for a very short span of time. However, the latest advancements in neuroscience, such as fluorescence imaging, have enabled the recording of responses of a large assembly of neurons for a longer span of time, which renders it possible to simulate the train of spikes emanating from a neuron when it is triggered (Shuttleworth et al. 2003). Following this line of research, the Spiking Neural Network (SNN) was developed that represents information as a train of spikes (Taherkhani et al. 2020). The input information is encoded into the relative timing between spikes. A neuron generates a spike when the membrane

Table 1 Latest progress in the development of biologically plausible neural networks for computer vision based on unsupervised learning

Authors	Biologically plausible neural network developed	Biological connect
Wallis and Rolls, (1997); Rolls et al. (2018)	VisNet: a four-layered hierarchical neural network for view-invariant object recognition that learns by competitive learning in an unsupervised manner	Only four stages in the architecture corresponding to 4–5 cortical areas in the ventral visual stream
		Network utilizes a temporal trace rule to maximize the agreement between invariant views that occur close together in time
		Slow unsupervised learning with convergence from stage to stage as in the ventral cortical visual stream
		Local synaptic rule is used for weight adaptation instead of backpropagation in deep neural networks
Zhuang et al. (2021)	Unsupervised model of the ventral visual stream based on contrastive learning	Emulates the manner in which infants learn from visual stimuli in a self-supervised manner from very few examples to learn from
		Brain-like representations are produced even when the visual stimuli are noisy
		Advocated the use of contrastive learning procedures such as SimCLR for emulating neuronal responses in the ventral visual stream
		Contrastive learning generates image embeddings that are viewpoint invariant just as in the primate brain
		The generated embeddings are consistent with the V1, V4 and ITC neuronal responses
Higgins et al. (2021)	Beta-VAE: a deep unsupervised model based on disentanglement representation learning	Disentangles sensory data into interpretable latent factors such as gender and age
		Disentangling presented as a biologically plausible objective function
		The disentanglement representations were found consistent with ITC neuronal responses to faces in macaque monkeys
		A Variational Auto-Encoder (VAE) is used as the deep unsupervised model
		The model was trained on 2100 mirror-flipped faces presented as visual stimuli to primates
Bakhtiari et al. (2021)	Contrastive predictive coding network that self-organizes into two parallel pathways	Simulates the ventral visual stream for recognizing the objects and the dorsal visual stream for localizing the object
		Self-supervised learning used to train two separate pathways
Mineault et al. (2021)	Self-supervised learning procedure to estimate the motion parameters of an agent	Simulates the sensitivity of the dorsal visual stream to motion
		The parameters of agent's motion estimated form the vision modality (multimodal learning)

Table 1 (continued)

Authors	Biologically plausible neural network developed	Biological connect
		The method captured the direction-oriented responses of the neuronal cells in the dorsal visual pathway
Li et al. (2022)	Consistency and complementarity network (CoCoNet): a self-supervised learning network	Generates multi-view representations of the object
		Learns through complementarity-aware contrastive learning approach
		Local complementarity model captures unique features of a view
		Global complementarity model captures global features shared across all views
		Uses both the ventral and dorsal visual streams to obtain multiple views of the object
Tang et al. (2022)	Biological Neural Networks (BNN) that operate by self-supervised learning	Biologically plausible alternative to backpropagation
		Self-supervised learning utilizes embeddings from unlabeled data to train the neural network
		Aim is to maximize the agreement between two different views of a related input, akin to human perception
		Biologically consistent due to the presence of direct connections between retinotopic layers and higher cortical areas (Van Essen et al. 2019)

potential exceeds a threshold (Holker and Susan 2021). The spiking neuron has more inherent properties that resemble a biological neuron than the perceptron. The SNN encodes images by converting pixel values into a train of spikes (Meftah et al. 2013). In the work of Cao et al. in 2015, SNNs were applied for computer vision tasks by mapping a trained CNN architecture to the SNN architecture (Cao et al. 2015). Recent advances in computing machines have made way for the development of large-scale spiking neural network based architectures, such as the SpiNNaker which is a part of the European Union's Human Brain project (Furber et al. 2014). Supervised training of multi-layer SNNs for high-end visual recognition tasks, to achieve high accuracies at par with deep learning, is a future possibility that may yield more biologically plausible computer vision models (Lagani et al. 2023; Tavanaei et al. 2019). Possibilities include conversion of pre-trained DNN to SNN, backpropagation derivatives for SNNs such as spike-based backpropagation approach and surrogate gradient learning, and unsupervised Spike Time Dependent Plasticity (STDP) learning rule (Bi and Poo 1998; Sjöström et al. 2001). Kheradpisheh et al. (2018) proposed a STDP-based deep SNN model for object recognition that comprises of trainable convolution layers (with unsupervised STDP), along with pooling layers. Due to STDP, neurons learned progressively from salient and frequent features in a hierarchical setup, and a higher-order neuron was used to ultimately predict the object category. The prime highlight of this unsupervised approach was that no class label was required for training, and only a few examples per object category were required. The authors justify their approach by drawing comparisons with real-world scenarios where the number of labeled samples is less, and assert that the learning process in humans and primates is mostly unsupervised.

Many software packages and neurosimulators have been developed for the computational study of brain networks. They support a large range of neurons and synapses. Examples are NEURON, GENESIS, BRIAN, NEST and ANNarchy, with NEURON showing the best performance, while BRIAN takes up more simulation time than the other neurosimulators (Tikidji-Hamburyan et al. 2017). While the hardware is still specialized and limited in its functionality, neurosimulators ease and accelerate the simulation of biologically plausible neural networks, similarly to PyTorch or Tensorflow for DNNs, although they have their own limitations for applications on multi-core computers or cluster computing.

Figure 4 summarizes the major breakthroughs in the evolution of biologically-inspired computer vision models over

Fig. 4 Major breakthroughs in the evolution and development of biologically-inspired computer vision models

time, as a conclusion of this study. The distinctive approaches that evolved over time, as highlighted in this study, are mentioned. The biological inspiration is evaluated from the perspective of resemblance in structural and/or functional behavior to the biological visual system.

4 Challenges and future directions

Deep convolutional neural networks demonstrate hierarchical signal processing that agrees with the progressive simple-to-complex signal processing in the biological visual system (Krüger et al. 2012). However, the similarity ends here; the biological visual system is much more complex than the present state of neural networks since the parallel streams of visual information intermix and exchange information at successive cortical stages of the brain. Unlike the biological network connections that includes reciprocal and parallel connections at every stage (Felleman and Van Essen 1991; Wyatte et al. 2014), the deep neural networks, in their present form, facilitate one-way feedforward hierarchy of visual signal processing. This leaves room for further improvement in the design of deep CNNs if they are required to achieve the perceptual and cognitive levels of animal or human vision. Simulation of this complex information exchange at various levels of the hierarchy is an open research problem for future vision systems. The development of recurrent convolutional neural networks (Spoerer et al. 2017) is a positive step in this direction. Integration of the top-down and bottom-up visual information by recurrent feedback connections in biological vision (Gilbert and Li 2013) motivates the development of computer vision models on similar lines, which ensures that the sensory information is supplemented by high-level visual information such as the visual attention map. Such integrative approaches are highly helpful in object segregation and identification in a crowded scene. A computational model based on this "reverse hierarchy" where the top-down knowledge was integrated with the sensory information was developed by Lee and Mumford (2003). CapsNets with its recurrent or feedback connections are projected by Horikawa and Kamitani (2017) to be a reliable model of biological vision in

the specific tasks of object segmentation and grouping. CapsNets have also proved successful in retaining information on object shape and geometry, unlike CNNs that cannot differentiate between possible and impossible geometric shapes as proved by Heinke et al. (2021). Further research on SNNs and CapsNets needs to be done to explore the possibility of their expansion to the scale of research done in CNNs. Hybrid models that incorporate the convolutional layers of CNNs and the spiking activity of SNNs with novel training methodologies may possibly yield more biologically plausible models. Medathati et al. (2016) gave indications of dynamic recurrent connections occurring in the visual pathway that allows the visual signal to bypass some of the cortical regions as it advances from the lower visual areas to the higher visual areas. Recent neuroimaging studies (Sugiura et al. 2020) have revealed direct transfer of neural activity between cortico-cortical areas of the posterior visual areas of the human brain. Simulation of this phenomenon would result in faster models for task-specific computer vision. Another advancement in deep learning is the integration of multiple-channel CNNs for processing parallel inputs from multiple sensors simultaneously (Palvanov et al. 2019). The training process of this integrated architecture is, however, time-consuming. The availability of large benchmark datasets, along with high performance machines, has ensured human-like performance of deep CNNs, albeit after weeks of training.

Some open research problems are summarized next, highlighting the research gaps in the translation of biological vision to computer vision. Some of these problems have been addressed by researchers but with none too satisfactory results.

I. The first challenge is the development of robust biologically-inspired models that are resilient to noise, that may take the form of background clutter, and at the same time also show transformation invariance. One technique for handling noisy data, inspired from the visual processing in the cortex, is local feature computation followed by integration by max-pooling which is equivalent to recording the maximum response of local receptive fields. This method which became popularly known as HMAX (Riesenhuber and Poggio 1999) is also invariant under translation, rotation, scale and viewpoint transformations, which is another test for robustness.
II. There is still a gap between human errors and machine errors (Geirhos et al. 2020). A lot depends on the size and variety of the dataset used to train the model to achieve human–machine consistency. Some factors that may contribute to reducing the gap include recurrent processing and feedback mechanisms (Kar and DiCarlo 2021; Kreiman and Serre 2020), and inductive bias for DNNs that implies adopting different training objectives for larger datasets and out-of-distribution datasets (Richards et al. 2019; Evans et al. 2022).
III. There are very few neural and behavioral benchmarks available to evaluate how close the computer vision model is to a biological visual system. One of the rare benchmarks available is BrainScore (Schrimpf et al. 2018) that evaluates how close an ANN is to a biological neural network for the object recognition task. The authors who developed BrainScore applied it to test a wide range of computer vision models, and concluded from their studies that DenseNet-169, CORnet-S and ResNet-101 are the best state-of-the-art computer vision models that approximate biological vision.
IV. One generally acknowledged problem with deep neural networks is the lack of explainability in its working (Blazek and Lin 2021). Though deep neural networks are getting to be as good as humans to recognize objects and scenes, the process by which it does so, is not very clear. The need to explain the functioning and inferencing performed by deep learning "black boxes" has given rise to a new research area called explainable AI (XAI) (Sattarzadeh et al. 2021) which can be done either "post-hoc" or "on-the-fly". Researchers currently working in XAI seek to decipher brain architecture and functioning using the computational model of deep neural networks. Blazek and Lin (2021) propose explainable neural networks called Essence Neural Networks (ENN) that can simulate human cognitive functions such as reasoning and out of distribution generalization. ENNs learn subconcepts by hierarchical clustering; subconcept nodes are connected to other nodes via direct or skip connections creating a hierarchical organization of concepts. Future research in explainable AI is expected to fill the gaps in the current computational models and the complex network structure in the brain.
V. There is inadequate research on spatio-temporal vision models that process temporal and spatial coordinates simultaneously for studying motion patterns of specific objects whose appearance cues are eligible. Rideaux and Welchman (2021) developed a neural network MotionNet_{xy} for estimating velocity using a regression layer, loosely based on the phenomenon of spatiotemporal information integration from the Middle Temporal (MT) and the V1 cortical region. The authors demonstrated that MT units tuned to slow speeds are integrated with V1 units tuned to low temporal frequency and high spatial frequency. Another achievement is the modeling of adaptive neuronal responses of continuous attractor neural network (CANN) found in the brain that can be applied for object tracking in real-world applications (Zou et al. 2022). Lin et al. (2021) proposed a reservoir decision-making network (RDMN) to mimic the

retina-subcortical pathway that integrates spatiotemporal signals for spatiotemporal pattern recognition such as gait recognition. The hierarchical reservoirs operate at different time and frequency scales which makes the spatiotemporal data linearly separable at a higher dimensional neural state space

VI. One problem with current deep networks emulating biological models is that they emulate the functioning of the brain by parts, and not as a whole. One possible research direction would be to integrate various specialized multi-tasking Deep CNNs to take a combined decision, as is done by the different cortical regions in the brain. Recent works on irregular deep vision models that consider locally connected receptive fields indicate a change in this direction (Nakada et al. 2021). A total of 24 deep neural networks and 4 shallow neural networks were used in (Nakada et al. 2021) to simulate artificial vision while perceiving a human in motion. New computational solutions that aid large-scale data analysis and management, such as increased computational capabilities with GPUs, cloud computing and distributed computing, can be employed for deploying such systems.

5 Conclusion

This paper highlights some important neuroscientific principles that may inspire the development of more efficient, adaptive and robust computer vision models. The progress achieved over the years in understanding biological vision and simulating parts of it with constrained success has been chronicled through a comprehensive set of works spanning over several decades. The hierarchical processing of signals in the visual pathway of the animal or human brain was found to inspire the feedforward hierarchy embedded in several state-of-the-art computer vision models like the neocognitron and CNNs. It was noted that as artificial neural networks evolved into more complex architectures over time, the biological connection was broken. However, continual studies in understanding the biological visual systems of different species spurred the development of biologically-plausible neural networks which is a highlight of this study. These include bio-inspired neural network architectures and unsupervised learning mechanisms that learn by self-supervision from unlabeled examples. Of special mention are the latest crop of unsupervised models that learn by self-supervision such as VisNet, CoCoNet, and Beta-VAE, that emulate the manner in which a child's brain develops over time. Contrastive learning procedures such as SimCLR and contrastive-learning models such as CLIP were found to be highly successful in emulating human vision. SNNs and CapsNets provide viable alternatives to CNNs, with unsupervised STDP-based SNN achieving object recognition capabilities compared to the biological visual system. To summarize, this study traces the biological connect of several state-of-the art computer vision models, and provides a detailed examination of the architectural design and learning mechanism of biologically-plausible neural networks. The findings suggest that the biological visual and cognitive system will influence, to a great extent, the development of future artificial vision systems.

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